# VARDHAMAN COLLEGE OF ENGINEERING
## DEPARTMENT OF CSE

Mini Project Review – II

**Title**: KhelQuest : AI–Powered Platform for Democratizing Sports.

Team No: H10

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Guide Details

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# KhelQuest: AI–Powered Platform for Democratizing Sports

- Project Title: KhelQuest: AI–Powered Platform for Democratizing Sports.
- Domain/Area: Artificial Intelligence (AI), Machine Learning (ML), Sports Analytics.
- Type: Application-Based Project.
- Keywords: AI in Sports, Talent Identification, Smartphone-based Analysis, Computer Vision, Cloud Analytics.

# BACKGROUND RESEARCH

- Existing Systems:

  Traditional sports talent identification systems:
  - Depend on expensive equipment like force plates, sensors, and motion capture systems.
  - Require trained professionals and specialized training centers.
  - Are mostly available in urban areas.
  - Use manual evaluation, which is time-consuming and biased.
- Gaps and Challenges:
  - Lack of equal opportunities for all athletes.
  - High cost of equipment and infrastructure.
  - Human bias in evaluation.
  - No real-time feedback for athletes.
  - Lack of secure and verified performance data.

- Motivation of the Project:

  The motivation behind KhelQuest is:

  - To democratize sports talent identification.
  - To use AI and smartphones to replace costly equipment.
  - To ensure transparent and unbiased evaluation.
  - To help athletes improve performance through instant feedback.
  - To support India’s vision of nurturing sports talent.

# PROJECT PLANNING

- **Research Phase**
  - Study AI in sports and computer vision techniques.
  - Analyze existing systems and technologies.
  - Review literature related to pose estimation and video analysis.
- **Design Phase**
  - Design system architecture.
  - Select AI models and cloud platform.
  - Design user interface and workflow.
- **Implementation Phase**
  - Develop mobile application for athletes.
  - Implement AI-based movement analysis.
  - Integrate cloud storage and dashboards.
- **Testing Phase**
  - Test accuracy of AI models.
  - Validate performance evaluation.
  - Check usability on low-end devices and low connectivity.

# Project Planning

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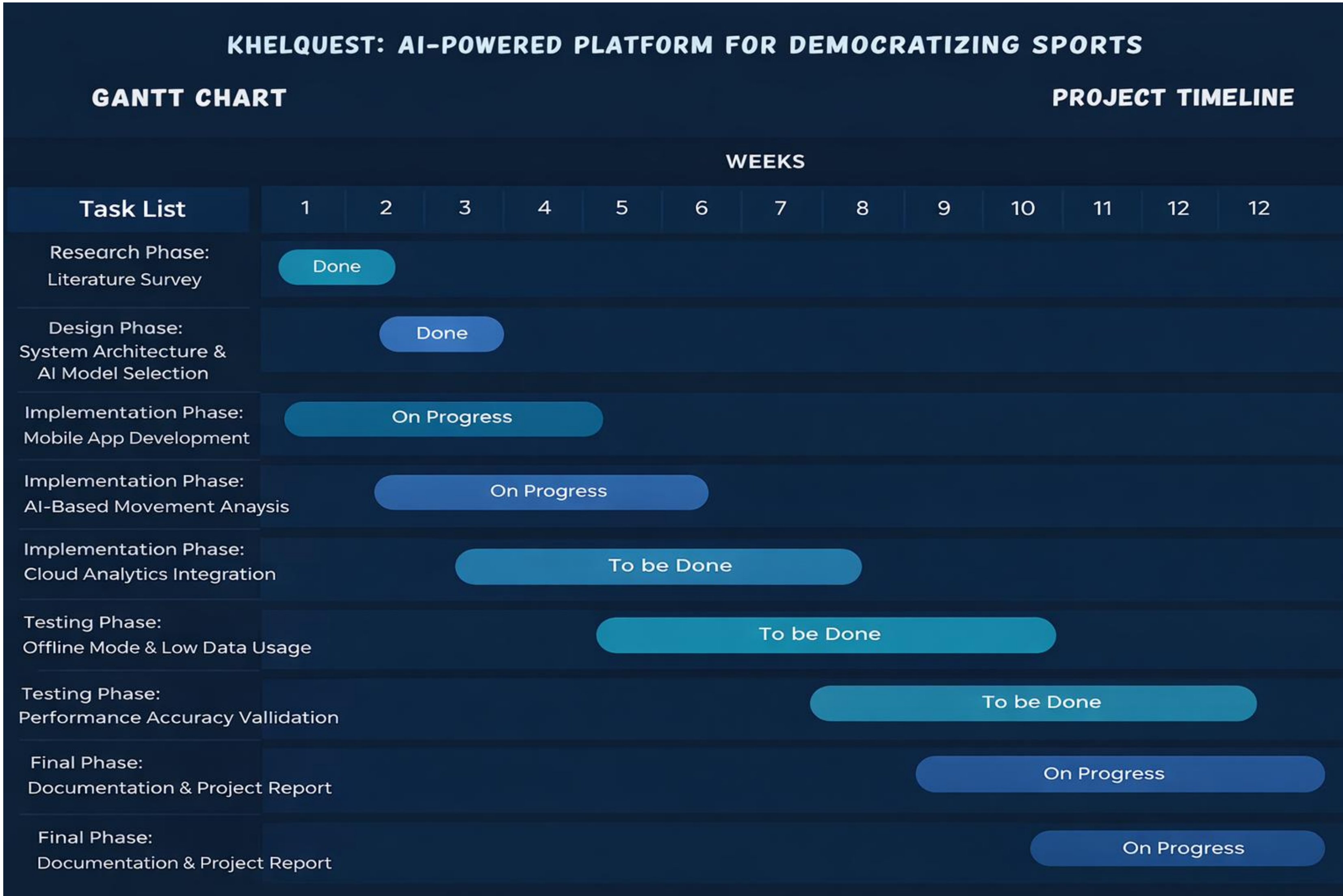


# LITERATURE SURVEY

| Paper Title | Approach/Algorithm | Dataset Used | Key Findings | Gaps Identified |
|---|---|---|---|---|
| *A Review of Vision-Based Motion Analysis in Sport* (Barris & Button, 2008) | Computer vision–based motion tracking using video analysis | Sports activity video recordings | Vision-based systems can analyze athlete movement without physical sensors | High computational cost; not optimized for mobile or low-resource environments |
| *OpenPose: Realtime Multi-Person 2D Pose Estimation* (Cao et al., 2019) | CNN-based pose estimation using keypoint detection | COCO Dataset, MPII Dataset | Accurate real-time pose detection for multiple users | Requires high processing power; limited mobile deployment |

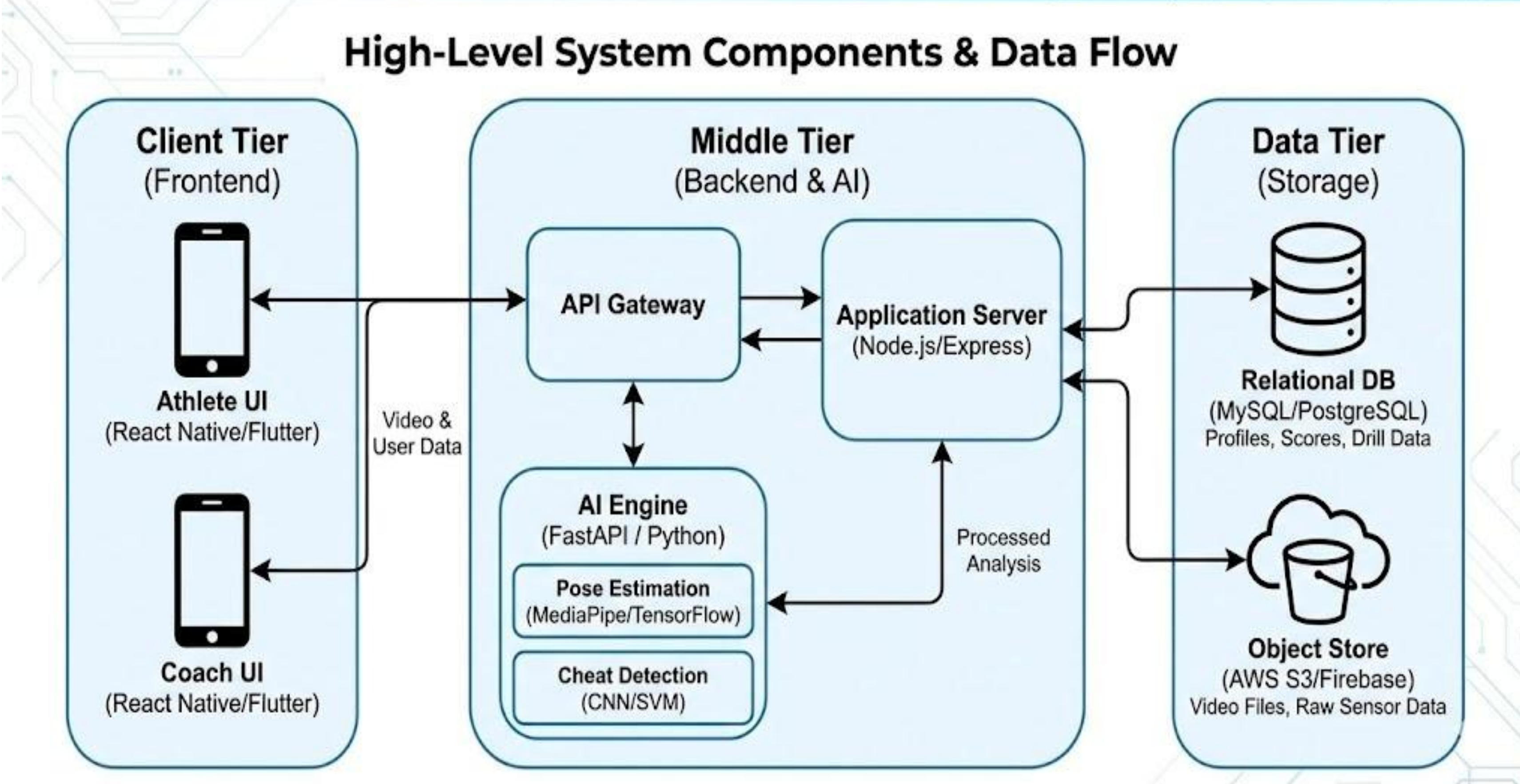
# LITERATURE SURVEY

| Paper Title | Approach/Algorithm | Dataset Used | Key Findings | Gaps Identified |
|---|---|---|---|---|
| *Machine Learning Applications in Sports Performance Analysis* (IEEE Xplore) | ML models for performance prediction and analysis | Sports performance datasets | ML improves objectivity and reduces human bias | Mostly focused on elite athletes and controlled environments |
| *Analysis of Accuracy of Vertical Jump Using Low-Cost Systems* (2023) | Video-based jump height estimation | Vertical jump test videos | Smartphone-based systems can reliably measure jumps | Limited scalability and no centralized data management |

# OBJECTIVES

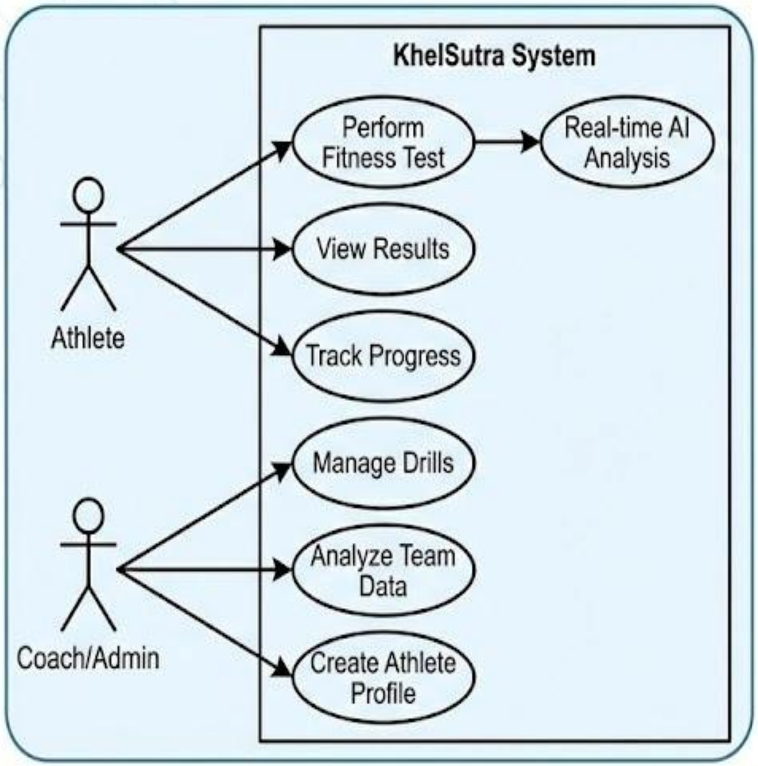
- To make sports talent identification accessible to all athletes.
- To replace expensive equipment with AI-based video analysis.
- To ensure fair and unbiased evaluation using AI.
- To provide instant feedback to athletes for performance improvement.
- To securely store athlete performance data in the cloud.
- To provide analytics and dashboards for coaches and sports authorities.
- To ensure the platform works in rural and low-connectivity areas.
- To contribute to India’s sports ecosystem and talent development.

# Design Phase: Project Architecture

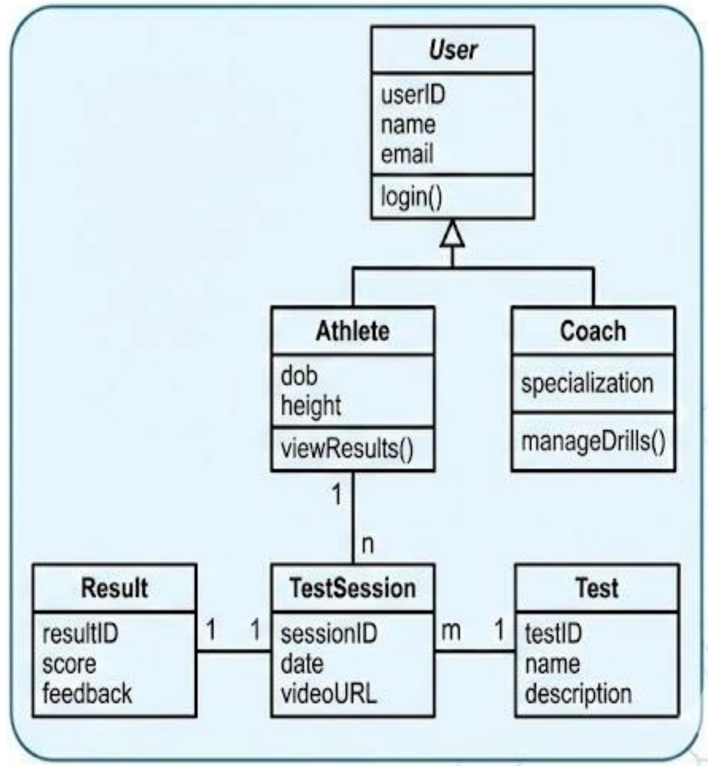


# System Modeling: Structure & Interactions

## Use Case Diagram (User Interactions)




## Class Diagram (Static Structure)



# PROJECT MILESTONE TRACKER

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# Q&A